

STAT536HW4

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2025-11-08

```
data <- matrix(c(28, 656,
                 18, 658),
              nrow = 2, byrow = TRUE,
              dimnames = list(Treatment = c("Placebo", "Aspirin"),
                              MI         = c("Yes", "No")))
data
```

```
##           MI
## Treatment Yes  No
## Placebo    28 656
## Aspirin    18 658
```

Q1

```
row_totals <- rowSums(data)
col_totals <- colSums(data)
grand_total <- sum(data)

row_totals; col_totals; grand_total
```

```
## Placebo Aspirin
##      684      676
```

```
## Yes    No
##  46 1314
```

```
## [1] 1360
```

Q2

```
r <- rowSums(data)
c <- colSums(data)
N <- sum(data)

bounds_2x2 <- function(r, c, N){
  L11 <- max(0, r[1] + c[1] - N); U11 <- min(r[1], c[1])
```

```

L12 <- max(0, r[1] + c[2] - N); U12 <- min(r[1], c[2])
L21 <- max(0, r[2] + c[1] - N); U21 <- min(r[2], c[1])
L22 <- max(0, r[2] + c[2] - N); U22 <- min(r[2], c[2])
data.frame(cell = c("a11(Placebo-Yes)", "a12(Placebo-No)",
                    "a21(Aspirin-Yes)", "a22(Aspirin-No)"),
           lower = c(L11, L12, L21, L22),
           upper = c(U11, U12, U21, U22))
}
bounds_2x2(r, c, N)

```

```

##           cell lower upper
## 1 a11(Placebo-Yes)      0    46
## 2 a12(Placebo-No)   638   684
## 3 a21(Aspirin-Yes)      0    46
## 4 a22(Aspirin-No)   630   676

```

This means the first cell, which reflects to Placebo-Yes, can vary from 0 to 46 while keeping all margins constant, and the other cells adjust accordingly. Therefore, there are 47 valid tables that satisfy the given row and column totals.

Q3

```

L11 <- max(0, r[1] + c[1] - N); U11 <- min(r[1], c[1])

num_tables <- U11 - L11 + 1
num_tables

```

```
## [1] 47
```

```

all_tables <- data.frame(
  a11 = L11:U11,
  a12 = r[1] - (L11:U11),
  a21 = c[1] - (L11:U11),
  a22 = N - (L11:U11) - (r[1] - (L11:U11)) - (c[1] - (L11:U11))
)
head(all_tables)

```

```

##   a11 a12 a21 a22
## 1   0 684  46 630
## 2   1 683  45 631
## 3   2 682  44 632
## 4   3 681  43 633
## 5   4 680  42 634
## 6   5 679  41 635

```

Q4

```
expected_saturated <- data
expected_saturated
```

```
##           MI
## Treatment Yes  No
##   Placebo  28 656
##   Aspirin  18 658
```

Under the saturated model, it is clear that the expected counts match the observed counts exactly.

Q5

```
chi_ind <- chisq.test(data, correct = FALSE)
expected_independence <- chi_ind$expected
expected_independence
```

```
##           MI
## Treatment      Yes      No
##   Placebo 23.13529 660.8647
##   Aspirin 22.86471 653.1353
```

Under the independence model, expected counts differ slightly from the observed values, reflecting a weak association.

Q6

```
test_res <- chisq.test(data, correct = FALSE)
test_res$statistic
```

```
## X-squared
##  2.129972
```

```
test_res$parameter
```

```
## df
##  1
```

```
test_res$p.value
```

```
## [1] 0.1444434
```

```
test_res
```

```
##
## Pearson's Chi-squared test
##
## data: data
## X-squared = 2.13, df = 1, p-value = 0.1444
```

The Pearson chi-square test indicates no significant relationship between aspirin use and myocardial infarction.

Q7

```
expected <- chisq.test(data, correct = FALSE)$expected
G2 <- 2 * sum(data * log(data / expected))
df <- (nrow(data) - 1) * (ncol(data) - 1)
p_value_G2 <- 1 - pchisq(G2, df)
p_value_G2
```

```
## [1] 0.1428159
```

The likelihood-ratio test also leads to the same as the Pearson chi-square test, indicating no significant relationship between aspirin use and myocardial infarction.

Q8

```
fisher_test <- fisher.test(data, alternative = "two.sided")
fisher_test
```

```
##
## Fisher's Exact Test for Count Data
##
## data: data
## p-value = 0.1768
## alternative hypothesis: true odds ratio is not equal to 1
## 95 percent confidence interval:
## 0.8236804 3.0256940
## sample estimates:
## odds ratio
## 1.559785
```

Fisher's exact test also fails to reject independence; the odds ratio confidence interval includes 1.

Q9

```
odds_yes <- data["Placebo","Yes"] / data["Aspirin","Yes"]
odds_no  <- data["Placebo","No"]  / data["Aspirin","No"]

log_ratio_yes <- log(odds_yes)
log_ratio_no  <- log(odds_no)

cat("log odds ratio given MI=Yes:", log_ratio_yes, "\n")
```

```
## log odds ratio given MI=Yes: 0.4418328
```

```
cat("log odds ratio given MI=No :", log_ratio_no, "\n")
```

```
## log odds ratio given MI=No : -0.003044142
```

The log-odds ratio shows that the placebo group has slightly higher MI odds.

Q10

Overall, all tests suggest that aspirin use is not significantly associated with myocardial infarction in this dataset.